

Course Outline: Summer-2025 (Section 15 and 16)

Course Title : Engineering Chemistry-I
 Course Code : CHE109
 Pre- requisite : None
 Class Time & Office Hour :

Time & Date	Course Code	Class Hour	Room No	Office time
Mon	CHE109 S-15 CHE109 S-16 CHE109 S-15	08:30 AM -10:00 AM 10:10 AM -11:40 AM 01:30 PM -03:30 PM	FUB-701 FUB-701 724 (Pharmacy Lab-1)	11.50 PM - 01.20 PM
Wed	CHE109 S-15 CHE109 S-16	08:30 AM -10:00 AM 10:10 AM -11:40 AM	FUB-701 FUB-701	11.50 PM - 01.20 PM

Make up Schedule : If necessary, any make-up class will be suitably arranged
 Course Instructor : **Prof. Dr. Md. Kamal Hossain (Adjunct Faculty at EWU)**
 Contact : **01911923724**
 Email : kamal.hossain@ewubd.edu
 Office : FUB 1004 (Dept. of MPS)

Course Objectives:

The learning objectives of this course are to promote fundamental knowledge in Engineering Chemistry and related fields to the students in order to

- understand the basic knowledge of atoms and molecules, elementary concepts of atomic structure, chemical bonding and shapes of molecules.
- convey knowledge on the mole concept and ideas of solution concentrations, acid-base titration and how structure relates to bonding and bulk properties.
- familiarize the students with the phases and solution behavior/properties, chemical equilibrium and thermodynamics, electrochemistry, nuclear chemistry, nano chemistry and the basics of organic and biochemistry.

Course Learning Outcomes: Upon completion of this course, the students will be able to

CLO1: Demonstrate and understand the basic knowledge of current chemical theories, chemical bonding (e.g. valence bond and VSEPR theories), equations, solution concentrations, mole calculations and to use these concepts in problem solving matters.

CLO2: Explain the basic analysis of acids and bases using different concepts. In addition, understand the chemical equilibria and thermochemistry including problem-solving matters.

CLO3: Demonstrate, design and carry out experiments with accuracy and analyze the results of such experiments. The students will be able to understand the different types of nuclear reactions.

CLO4: Understand the structures of bio-molecules, biochemistry of enzymes, membrane transport, metabolism of proteins, carbohydrates, lipids and nucleic acids.

Course Contents/Description:

Mid Exam

Atomic structure and properties: Atomic particles, atomic number, mass number, atomic orbitals, isotopes, electronic configurations of atoms, Pauli exclusion principle, Hund's rule, Aufbau principles atomic models/postulations, periodic table and periodic properties of elements, semiconductor, photoelectric effect, electromagnetic radiation.

Chemical solution of acids and bases: Types of solutions and ways of expressing concentration, Colligative properties, Strong and weak acids/bases, relative strength of acids/base, pH and buffer solutions, buffer capacity, neutralization curves, acid base titration, Measurement of pH.

Chemical bonding: Forces between atoms and molecules, Classification of solid based on different binding forces, ionic bond, covalent bond, metallic bond, hydrogen bond, hybridization, valence bond theory (VBT), valence-shell electron-pair repulsion (VSEPR) theory, geometry of molecules.

Nuclear chemistry: Radioactive elements, properties of alpha, beta and gamma rays, radioactive decay, calculation of half-life period of radioactive isotope, radioactive dating, nuclear reaction.

Nanochemistry: Nanomaterials, classification and properties of nanomaterials, synthesis and processing of nanomaterials, Characterization techniques of nanomaterials, nanowires, carbon nanotubes, graphene. Application of nanomaterials in energy storage and environmental issues (air, water, and fuel purification).

Final Exam

Properties of gases: Gas Laws (Boyles law, Charles's Law, combined gas law, Gay-Lussac law, ideal gas law, Dalton's law of partial pressure). Mathematical problems.

Electrochemistry: Oxidants and reductants, oxidation number, electrode, electrolyte, standard hydrogen electrode, electrolysis, Electrochemical cell, electromotive force, electrode reactions, reduction potential, the chemical series, electroplating, product deposition within the electrodes, Galvanic cell, Daniel cell, dry cell, Hydrogen fuel cell.

Environmental chemistry: Sources and effects of air pollution, acid rain, ozone depletion, water pollutants, soil pollution, greenhouse effect, prevention of environmental pollution. water and air purification via adsorption.

Basics of organic chemistry: Introduction of aliphatic and aromatic hydrocarbons, nomenclature of various organic compounds, synthesis of various hydrocarbons (alkane, alkene, alkyne, alcohol, acid, benzene, phenol, etc.), important organic reactions such as addition reaction, elimination reaction, substitution reaction, Friedel craft reactions, Selected organic compounds (phenol and its derivatives; alcohols and its derivatives).

Basics of biochemistry: Photosynthesis, carbohydrates (saccharides), protein (amino acids, peptide bond), lipids, fat/oil (saturated/unsaturated), nucleic acids, DNA, RNA, hemoglobin and myoglobin, Enzymes.

Basics of Chemical Hazard and Chemical Weapons Convention (CWC): Basics of chemical safety, explosive/hazardous chemicals, storage/handling chemicals, hazard communications, health risks of chemical exposer, personal protective equipment, chemical safety rules and regulations, Chemical weapons and their classification, dual use of chemicals, chemical threat.

Suggested Readings:

1. General Chemistry, 11th Ed, by Darrell D. Ebbing, Steven D. Gammon
2. Chemistry, 9th Ed by Steven S. Zumdahl, Susan A. Zumdahl
3. Chemistry, 13th Edition, by Raymond Chang and Jason Overby
4. Inorganic Chemistry, 3rd Ed., C. E. Housecroft and A. G. Sharpe, Pearson, USA.
5. Organic Chemistry, 12th Ed, by T. W. Graham Solomons, Craig B. Fryhle, Scott A. Snyder
6. Physical Chemistry, P. W. Atkins, W. H. Freeman & Co.
7. Introduction to Nuclear and Radiochemistry, G. Friedlander, J. W. Kennedy, E. S. Macias and
8. Environmental Chemistry, S. Manahan, Lewis Publisher.

Evaluation:

Quizzes	10%
Presentations/Viva/Special Quiz	10%
Attendance	5%
Mid Term Exam	25%
Term Final Exam	25%
Lab	25%

Grading Policy:

Marks (%)	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	D	2.00
less than 40%	F	0.00

Exam Schedule: Last day of classes, 04 September 2025

Exam	Date	
MID	Monday	4 August 2025
Final	Wednesday	10 September 2025

Special Instructions:

- Students are required to arrive in class on time and are expected to attend all classes and examinations.
- **An apron is mandatory for the Lab class.**
- **No section changes are allowed for regular classes or practical labs.**
- **No make-up lab is allowed without valid reason.**
- All mobile phones **MUST** be turned to switch off during class and exam periods.
- Plagiarism in any assignments/lab reports will not be allowed.
- The act of cheating in exams is a punishable offense and strict actions will be taken according to EWU rules.



Prof. Dr. Md. Kamal Hossain